

MAE 184 Flight Simulation Techniques
Homework #3 Due FRI 23 MAY 2025

Please submit final answers using the formatted "Quiz" on the Canvas course website. This worksheet is for reference only.

The first two problems require the JSBSim program and the Cessna 182 aircraft model that has been modified for MAE 184 (i.e. not the version from the FlightGear website). The MAE 184 model is available on Canvas and in the simulation lab.

Problem 3.1 (2 pts) L13

The table below provides data from the airplane manufacturer regarding engine power required for the Cessna 182 to maintain level flight at 6,000 ft altitude. Use the JSBSim trim script (c182_trim.xml) to find the level-flight equilibrium for each of the flight conditions listed in the table. Use the JSBSim trim results to create a plot with true airspeed [units: kts, nmi/hr] on the x-axis and engine brake horsepower [units: hp] on the y-axis. Show the manufacturer data as a line, and your JSBSim data as individual dots. Be sure to show a legend and axis labels on your graph. Submit your plot in BMP, JPG, or PNG format.

Manufacturer Data 182P Cruise Performance (weight = 2950 lb)

True Airspeed [kts]	Brake Horsepower [hp]	Fuel Flow [gal/hr]
117	113	9.4
122	122	10.0
127	131	10.6
131	140	11.3
135	147	11.9
138	156	12.6
141	166	13.3
144	175	14.0

Problem 3.2 (2 pts) L13, L18

Use the manufacturer fuel flow data from the previous problem to estimate the Brake Specific Fuel Consumption (BSFC) from the Cessna 182 model. Assume avgas fuel weighs 6.0 lbf/gal.

$$\frac{dW}{dt} = -c_p P \quad W = \text{weight}, P = \text{engine brake power}, t = \text{time}, c_p = \text{BSFC}$$

$$\text{BSFC}, c_p = \frac{0.48647}{0.48154 \text{ [JSBSim]}} \text{ [lbf/(hr.hp)]}$$

Problem 3.3 (2 pts) L15

Use 1D linear interpolation to estimate the lift coefficient (CL) for an angle-of-attack of 11.0 degrees.

x Angle-of-Attack (deg)	$f(x)$ Lift Coefficient (CL)
8.0	0.7130
12.0	0.9450
14.0	1.0470
15.0	1.1330

$$f(x) = f(x_0) + \frac{x - x_0}{x_1 - x_0} [f(x_1) - f(x_0)]$$

a. 0.771

b. 0.829

c. 0.887

d. 0.945

$$C_{L_{11^\circ}} = C_{L_{8^\circ}} + \frac{11^\circ - 8^\circ}{12^\circ - 8^\circ} [C_{L_{12^\circ}} - C_{L_{8^\circ}}]$$

$$= 0.7130 + \frac{3}{4} [0.9450 - 0.7130]$$

$$C_{L_{11^\circ}} = 0.887$$

Problem 3.4 (2 pts) L16

The fuel tanks of the Cessna 182 are located in the wings of the airplane. Estimate the body-axis moment of inertia I_{xx} contributed by the two fuel tanks. Assume the tanks are full of fuel and are shaped like a **rectangular prism** with dimensions given below.

fuel mass = 8.6 slugs (each tank)

fuel tank dimensions (each tank)

width (leading edge to trailing edge of wing) = 3 ft (w)

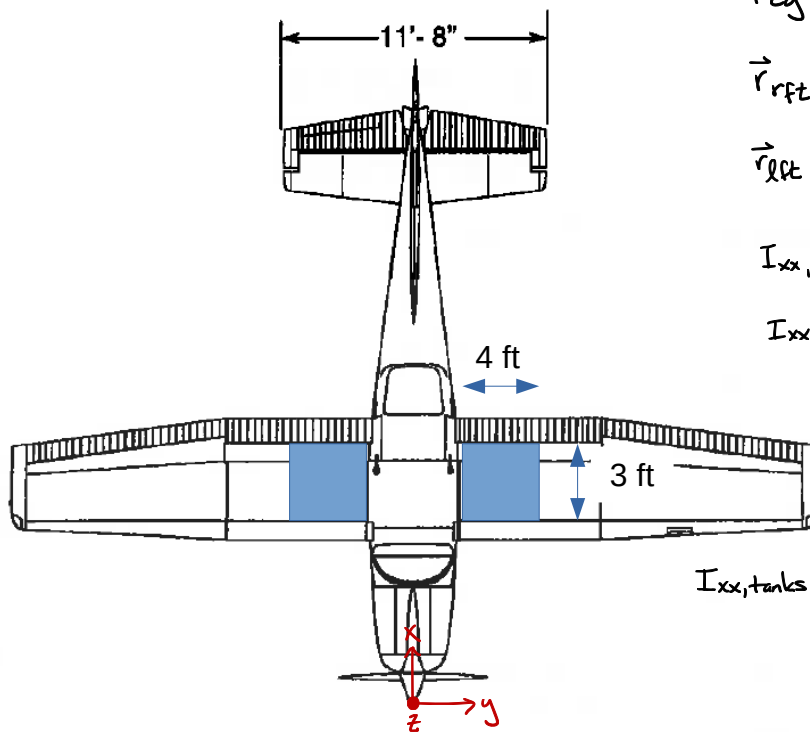
height (bottom of wing to top) = 0.5 ft (H)

length (root of wing toward tip) = 4 ft (L)

c.g. location in structural axis = (41.0, 0.0, 36.5) inches

right fuel tank center-of-mass in structural axis =
(48.0, 48.0, 59.4) inches

left fuel tank center-of-mass in structural axis =
(48.0, -48.0, 59.4) inches



$$\begin{aligned} \vec{r}_{cg} &= \left(\frac{41}{12}, 0, \frac{36.5}{12} \right) \text{ ft} \\ \vec{r}_{rft} &= (4, 4, 4.95) \text{ ft} \\ \vec{r}_{lft} &= (4, -4, 4.95) \text{ ft} \end{aligned} \quad \left. \begin{array}{l} \text{Displacement Mag.} \\ \Delta y_c = 4 \\ \Delta z_c = 1.9083 \end{array} \right\}$$

$$I_{xx, \text{tank}} = I_{xx, \text{centroid}} + m_c (\Delta y_c^2 + \Delta z_c^2)$$

$$I_{xx, \text{centroid}} = \frac{m}{12} (L^2 + H^2)$$

$$\begin{aligned} I_{xx, \text{tanks}} &= 2 \left(I_{xx, \text{centroid}} + m_c (\Delta y_c^2 + \Delta z_c^2) \right) \\ &= 2 \left(\frac{m_c}{12} (L^2 + H^2) + m_c (\Delta y_c^2 + \Delta z_c^2) \right) \\ &= 2 \left(\frac{8.6}{12} ((4)^2 + (0.5)^2) + 8.6 ((4)^2 + (1.9083)^2) \right) \end{aligned}$$

$$I_{xx} \text{ (both tanks full)} = \underline{361.1295} \text{ [slug.ft}^2\text{]}$$

$$I_{xx, \text{tanks}} = 361.1295 \text{ slug.ft}^2$$

Problem 3.5 (2 pts) L18

Use the propeller efficiency curve shown below to estimate the thrust force [units: lbf] that the propeller is generating if the motor produces 130 hp. The 6.75 ft diameter propeller is rotating at a rate of 1800 rev/min and the aircraft is traveling at a true airspeed of 170 ft/s. (note: 1 hp = 550 ft.lb/s)

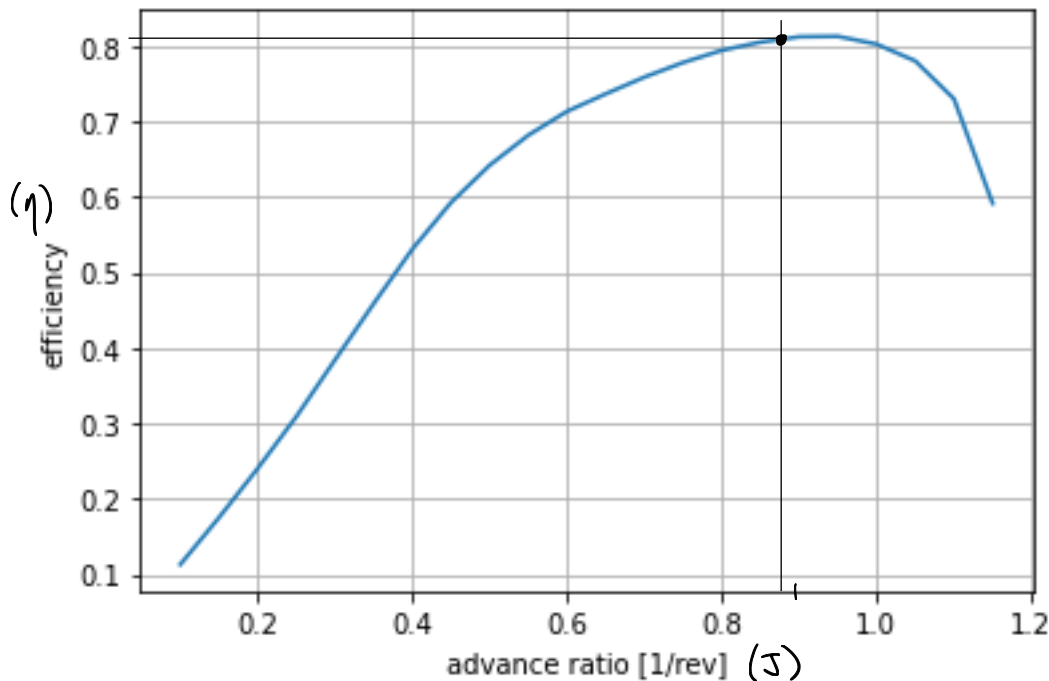
Thrust force = 340.6765 [lbf]

$$P = 130 \text{ hp} \rightarrow 71500 \frac{\text{ft} \cdot \text{lb}}{\text{s}}$$

$$D = 6.75 \text{ ft}$$

$$n = 1800 \frac{\text{rev}}{\text{min}} \rightarrow 30 \frac{\text{rev}}{\text{sec}}$$

$$V = 170 \text{ ft/s}$$



1) Find J

$$J = \frac{V}{nD}$$

$$= \frac{170 \text{ ft/s}}{(30 \frac{\text{rev}}{\text{s}})(6.75 \text{ ft})}$$

$$J \approx 0.84 \frac{1}{\text{rev}}$$

2) Find corresponding η

$$\eta(0.84 \frac{1}{\text{rev}}) \approx 0.81 \text{ (from graph)}$$

3) Find T

$$\eta = \frac{TV}{P} \rightarrow T = \frac{\eta P}{V}$$

$$= \frac{(0.81)(71500 \frac{\text{ft} \cdot \text{lb}}{\text{s}})}{170 \frac{\text{ft}}{\text{s}}}$$

$$T \approx 340.6765 \text{ lbf}$$